



**Michigan
Technological
University**

Mechanical and Aerospace Engineering

Calibration Certificates

Declan Jordan - 25% Sierra Manriquez - 25% John Dombrowski - 25% Andrew Shepard - 25%

L14 Team 2

Calibration Certificate

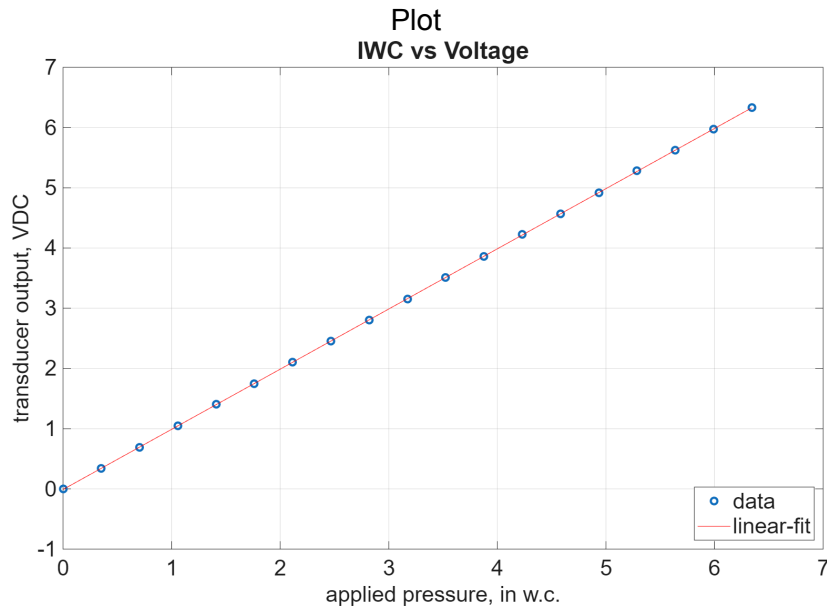
manufacturer: BAPI
model number: Ba/zpm/hr-nt-nd-bb
serial number: PT2A
calibration range: 0 to 10 in w.c.
output signal: 0 to 10 VDC
accuracy: $\pm 0.25\%$ F.S.

date: 1/29/2026
time: 04:15 pm
section: L14

technicians: John Dombrowski
Declan Jordan
Sierra Manriquez
Andrew Shepard

calibration station: Inclined Manometer
indicator oil SG: 0.881
Hg SG: 13.55
pressure port: 1

Pressure in w.c.	Output VDC
0.00	0.000
0.40	0.341
0.80	0.691
1.20	1.045
1.60	1.405
2.00	1.745
2.40	2.101
2.80	2.455
3.20	2.805
3.60	3.155
4.00	3.511
4.40	3.857
4.80	4.223
5.20	4.565
5.60	4.921
6.00	5.281
6.40	5.621
6.80	5.975
7.20	6.331



sensitivity, y/x:	0.9992	VDC per in w.c.
bias, y:	-0.0098	VDC
R ² :	1.0000	
sensitivity, x/y:	1.0008	in w.c. per VDC
bias, x:	0.0098	in w.c.

Calibration Certificate

manufacturer: BAPI
model number: Ba/zpm/hr-nt-nd-bb
serial number: PT2A
calibration range: 0 to 30 in w.c.
output signal: 0 to 10 VDC
accuracy: $\pm 0.25\%$ F.S.

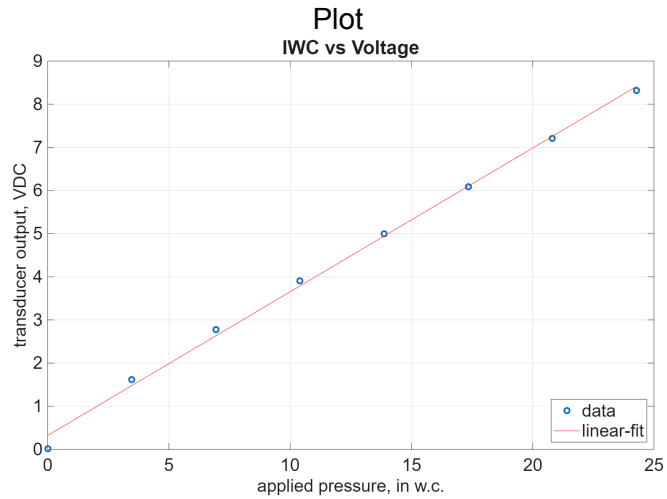
date: 1/29/2026
time: 04:30 pm
section: L14

calibration station: Dual U-tube Manometer Bench

indicator oil SG: 0.881
Hg SG: 13.55
pressure port: 1

technicians: John Dombrowski
Declan Jordan
Sierra Manriquez
Andrew Shepard

Pressure in w.c.	Output VDC
0.00	0.000
2.34	0.780
4.62	1.545
6.90	2.295
9.15	3.045
11.43	3.825
13.71	4.555
15.99	5.325
18.27	6.085
20.54	6.845
22.76	7.595



sensitivity, y/x:	0.3129	VDC per in w.c.
bias, y:	-0.0012	VDC
R ² :	1.0000	
sensitivity, x/y:	3.1956	in w.c. per VDC
bias, x:	0.0037	in w.c.

Calibration Certificate

manufacturer: BAPI
model number: Ba/zpm/hr-nt-nd-bb
serial number: 2B
calibration range: 0 to 10 in w.c.
output signal: 0 to 10 VDC
accuracy: $\pm 0.25\%$ F.S.

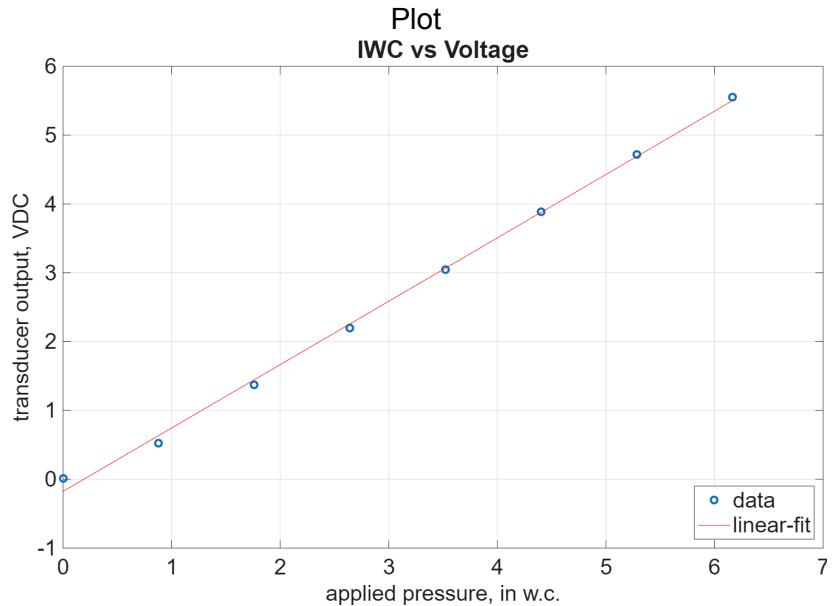
date: 1/29/2026
time: 04:50 pm
section: L14

technicians: John Dombrowski
Declan Jordan
Sierra Manriquez
Andrew Shepard

calibration station: Inclined Manometer

indicator oil SG: 0.881
Hg SG: 13.55
pressure port: 2

Pressure in w.c.	Output VDC
0.00	0.010
0.88	0.525
1.76	1.375
2.64	2.200
3.52	3.044
4.41	3.889
5.29	7.420
6.17	5.550



sensitivity, y/x:	0.9208	VDC per in w.c.
bias, y:	-0.1751	VDC
R ² :	0.9979	
sensitivity, x/y:	1.0860	in w.c. per VDC
bias, x:	0.1901	in w.c.

Calibration Certificate

manufacturer: BAPI
model number: Ba/zpm/hr-nt-nd-bb
serial number: 2B
calibration range: 0 to 30 in w.c.
output signal: 0 to 10 VDC
accuracy: $\pm 0.25\%$ F.S.

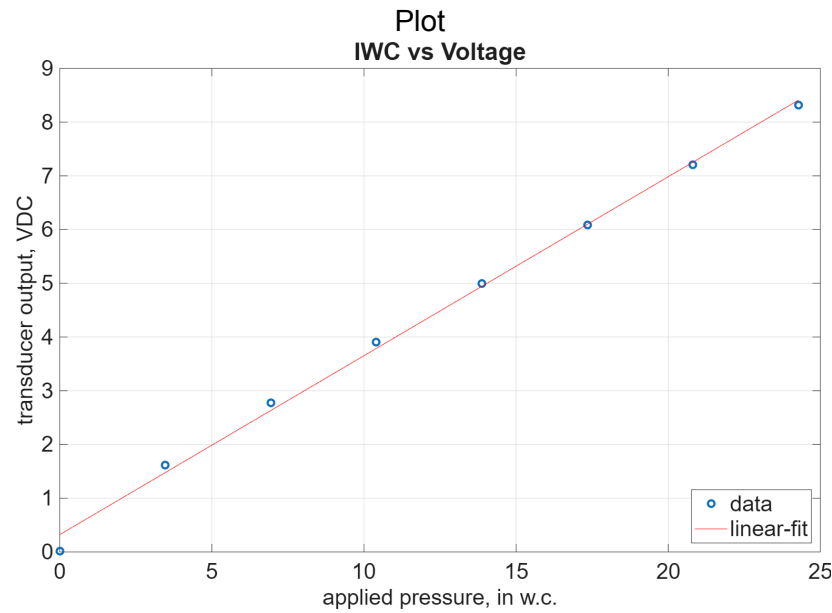
date: 1/29/2026
time: 05:10 pm
section: L14

technicians: John Dombrowski
Declan Jordan
Sierra Manriquez
Andrew Shepard

calibration station: U-Tube Manometer

indicator oil SG: 0.827
Hg SG: 13.55
pressure port: 2

Pressure in w.c.	Output VDC
0.00	0.010
3.26	1.621
6.51	2.779
9.77	3.910
13.02	5.000
16.28	6.089
19.54	7.207
22.79	8.313



sensitivity, y/x:	0.3332	VDC per in w.c.
bias, y:	0.3215	VDC
R ² :	0.9970	
sensitivity, x/y:	3.0015	in w.c. per VDC
bias, x:	-0.9650	in w.c.

Calibration Certificate

manufacturer: BAPI
model number: Ba/zpm/hr-nt-nd-bb
serial number: SPARE 1
calibration range: 0 to 10 in w.c.
output signal: 0 to 10 VDC
accuracy: $\pm 0.25\%$ F.S.

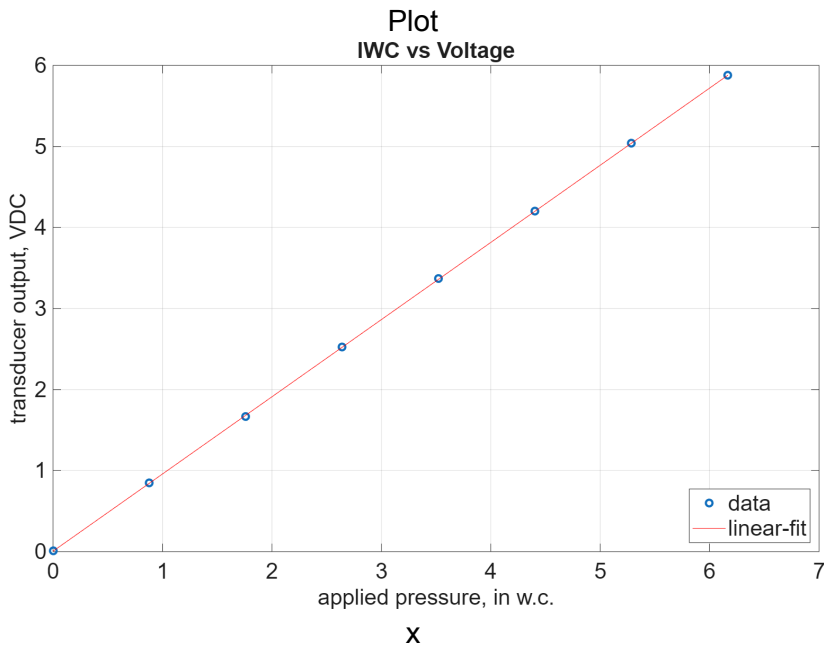
date: 1/29/2026
time: 05:30 pm
section: L14

technicians: John Dombrowski
Declan Jordan
Sierra Manriquez
Andrew Shepard

calibration station: Inclined Manometer

indicator oil SG: 0.881
Hg SG: 13.55
pressure port: 2

Pressure in w.c.	Output VDC
0.00	0.010
0.88	0.845
1.76	1.668
2.64	2.528
3.52	3.372
4.41	4.200
5.29	5.038
6.17	5.880



sensitivity, y/x:	0.9526	VDC per in w.c.
bias, y:	0.0053	VDC
R ² :	1.0000	
sensitivity, x/y:	1.0498	in w.c. per VDC
bias, x:	-0.0056	in w.c.

Calibration Certificate

manufacturer: BAPI
model number: Ba/zpm/hr-nt-nd-bb
serial number: SPARE 1
calibration range: 0 to 30 in w.c.
output signal: 0 to 10 VDC
accuracy: $\pm 0.25\%$ F.S.

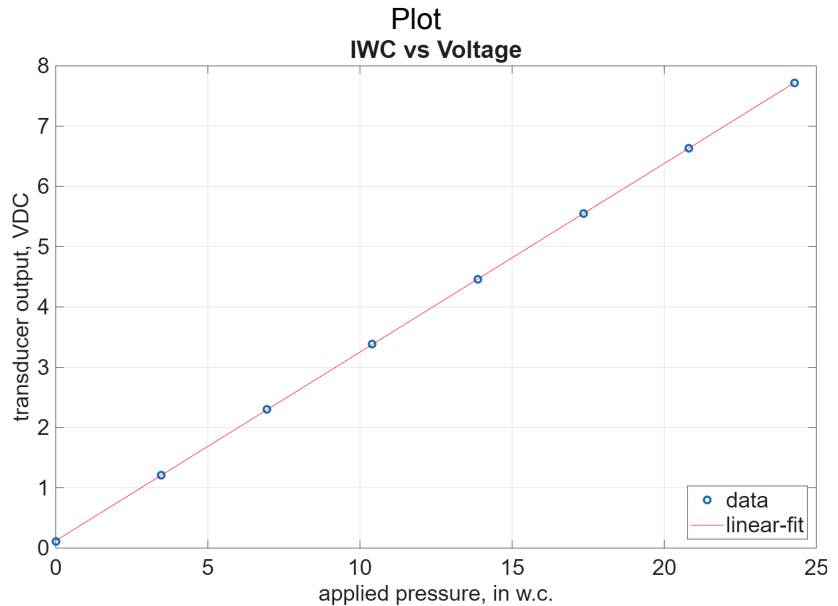
date: 1/29/2026
time: 05:50 pm
section: L14

calibration station: Inclined Manometer

indicator oil SG: 0.827
Hg SG: 13.55
pressure port: 2

technicians: John Dombrowski
Declan Jordan
Sierra Manriquez
Andrew Shepard

Pressure in w.c.	Output VDC
0.00	0.011
3.26	1.213
6.51	2.302
9.78	3.380
13.02	4.459
16.28	5.554
19.54	6.635
22.79	7.714



sensitivity, y/x:	0.3129	VDC per in w.c.
bias, y:	0.1220	VDC
R^2 :	1.0000	
sensitivity, x/y:	3.1956	in w.c. per VDC
bias, x:	-0.3899	in w.c.

Transducer Calibration Data Log-Low Range - PT #A
Transducer Information

Manufacturer	BAPI
Transducer Model:	Ba/zpm/hr-nt-nd-bb
Transducer Number:	PT2A
Transducer Range:	0 - 10 in W.C.
Output Signal	0-10 VDC
Accuracy	Thousandths

Calibration Station-Manometer as Standard

Pressure Port:	1
Manometer Type:	Inclined
Specific Gravity:	0.881
Calibration Scale:	inches oil
Calibration Uncertainty:	0.005

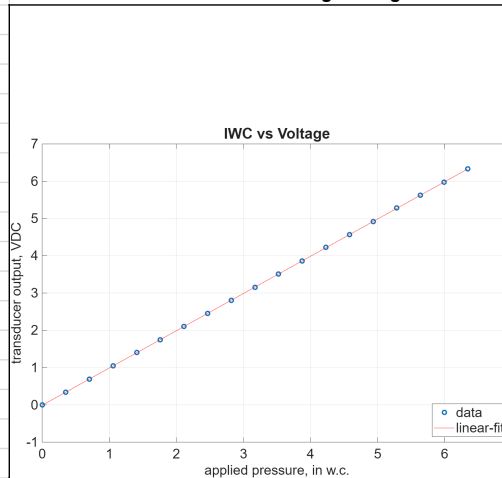
Test Point	Manometer scale in oil	Output VDC
1	0	0
2	0.4	0.341
3	0.8	0.691
4	1.2	1.045
5	1.6	1.405
6	2	1.745
7	2.4	2.101
8	2.8	2.455
9	3.2	2.805
10	3.6	3.155
11	4	3.511
12	4.4	3.857
13	4.8	4.223
14	5.2	4.565
15	5.6	4.921
16	6	5.281
17	6.4	5.621
18	6.8	5.975
19	7.2	6.331

Sensitivity y/x:	0.9992	VDC per in. WC
Bias:	-0.0098	VDC
R^2	1	
Inverted Sensitivity x/y:	1.0008	in. WC per VDC

Technician Information

Date:	
Section:	
Team #:	
Members:	

DO NOT CHANGE/EDIT THE SHEET NAME

Observations and Comments During Testing


[illegible]

Transducer Calibration Data Log-High Range - PT #A

Transducer Information

Manufacturer	BAPI
Transducer Model:	Ba/zpm/hr-nt-nd-bb
Transducer Number:	PT2A
Transducer Range:	0 - 30
Output Signal	0-10
Accuracy	Thousandths

Technician Information

Date:	1/30/2026
Section:	L14
Team #:	T2
Members:	

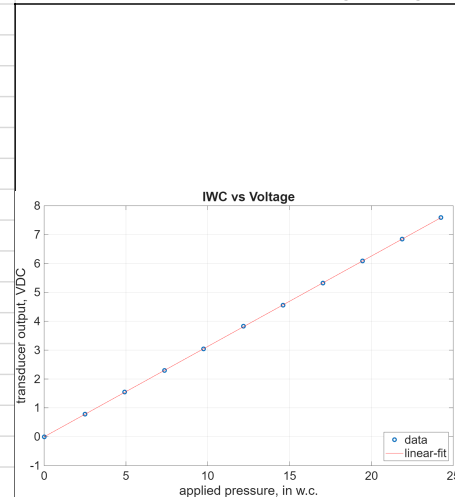
DO NOT CHANGE/EDIT THE SHEET NAME

Calibration Station-Manometer as Standard

Pressure Port:	1	
Manometer:	Oil	Mercury
Manometer Type:	u tube	u tube
Specific Gravity:	0.827	13.55
Calibration Scale:	mm Oil	mm Hg
Calibration Span:	800 mm	800 mm
Calibration Uncertainty:	0.5 mm	0.5 mm

Test Point	Manometer scale mm oil	Manometer scale mm Hg	Output VDC
1	0	0	0
2	72	1	0.78
3	142	4	1.545
4	212	12	2.295
5	281	17	3.045
6	351	22	3.825
7	421	27	4.555
8	491	31	5.325
9	561	35	6.085
10	631	39	6.845
11	699	44	7.595
12			
13			
14			
15			
16			
17			
18			
19			

Observations and Comments During Testing



[illegible]

Transducer Calibration Data Log-Low Range - PT #B

Transducer Information

Manufacturer	BAPI
Transducer Model:	BAZPM-HR-NT-ND-BB
Transducer Number:	2B
Transducer Range:	0 -10 in w.c.
Output Signal	0-10 VDC
Accuracy	Thousandths

Calibration Station-Manometer as Standard

Pressure Port:	2
Manometer Type:	Inclined
Specific Gravity:	0.881
Calibration Scale:	inches oil
Calibration Uncertainty:	

Test Point	Manometer scale in oil	Output VDC
1	0	0.01
2	1	0.525
3	2	1.375
4	3	2.2
5	4	3.044
6	5	3.889
7	6	4.72
8	7	5.55
9		
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		

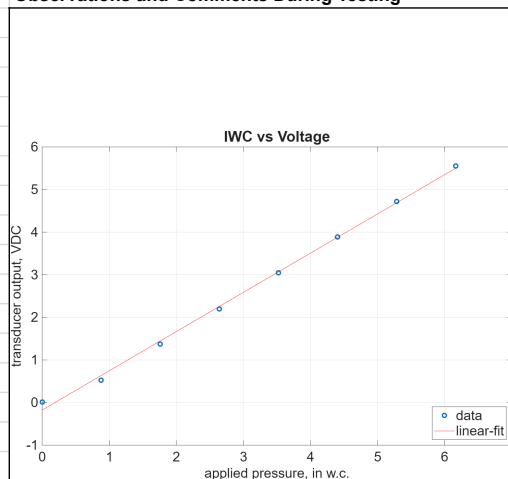
Sensitivity y/x:	0.9208	VDC per in. WC
Bias:	-0.1751	VDC
R^2	0.9979	
Inverted Sensitivity x/y:	1.086	in. WC per VDC

Technician Information

Date:	1/30/2026
Section:	L14
Team #:	T2
Members:	

DO NOT CHANGE/EDIT THE SHEET NAME

Observations and Comments During Testing



Inverted Bias:	0.1901	in. WC								
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Transducer Calibration Data Log-High Range - PT #B

Transducer Information

Manufacturer	BAPI
Transducer Model:	Ba/zpm/hr-nt-nd-bb
Transducer Number:	2B
Transducer Range:	30 in w.c.
Output Signal	0-10 VDC
Accuracy	Thousandths

Technician Information

Date:	1/30/2026
Section:	L14
Team #:	T2
Members:	

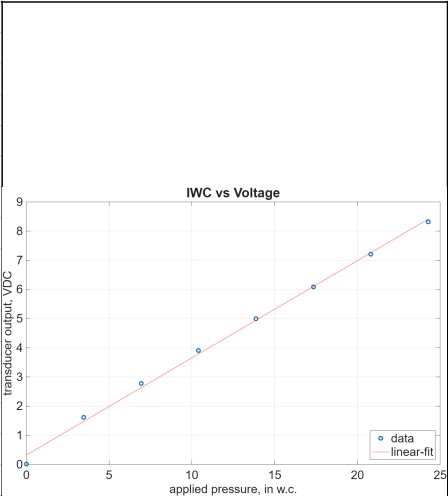
DO NOT CHANGE/EDIT THE SHEET NAME

Calibration Station-Manometer as Standard

Pressure Port:	2	
Manometer:	Oil	Mercury
Manometer Type:	U-Tube	U-Tube
Specific Gravity:	0.827	13.55
Calibration Scale:	mmOil	mmHg
Calibration Span:	700mm	42mm
Calibration Uncertainty:		

Test Point	Manometer scale mm oil	Manometer scale mm Hg	Output VDC
1	0	0	0.01
2	100	6	1.621
3	200	12	2.779
4	300	18	3.91
5	400	24	5
6	500	30	6.089
7	600	36	7.207
8	700	42	8.313
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			

Observations and Comments During Testing



[illegible]

Transducer Calibration Data Log-Low Range - Spare PT#

Transducer Information

Manufacturer	BAPI
Transducer Model:	BAZPM-HR-NT-ND-BB
Transducer Number:	Spare 1
Transducer Range:	30
Output Signal	0-10 VDC
Accuracy	

Calibration Station-Manometer as Standard

Pressure Port:	1
Manometer Type:	Inclined
Specific Gravity:	0.881
Calibration Scale:	inches oil
Calibration Uncertainty:	Thousandths

Test Point	Manometer scale in oil	Output VDC
1	0	0.01
2	1	0.845
3	2	1.668
4	3	2.528
5	4	3.372
6	5	4.2
7	6	5.038
8	7	5.88
9		
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		

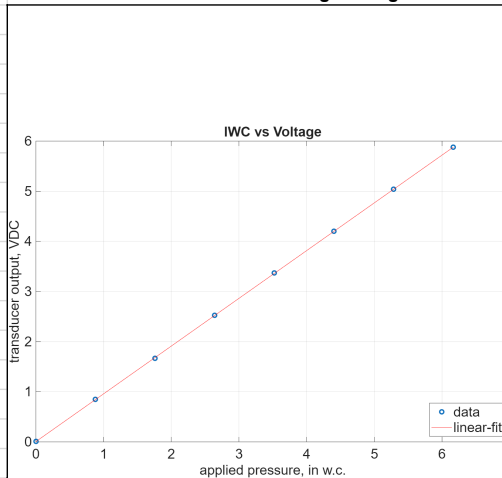
Sensitivity y/x:	0.9526	VDC per in. WC
Bias:	0.0053	VDC
R^2	1	
Inverted Sensitivity x/y:	1.0498	in. WC per VDC

Technician Information

Date:	1/30/2026
Section:	L14
Team #:	T2
Members:	

DO NOT CHANGE/EDIT THE SHEET NAME

Observations and Comments During Testing



Inverted Bias:	-0.0056	in. WC								
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Transducer Calibration Data Log-High Range - Spare PT #

Transducer Information

Manufacturer

BAPI

Transducer Model:

I BAZPM-HR-NT-ND-BB

Transducer Number:

Spare 1

Transducer Range:

0-30

in w.c.

Output Signal

0-10

VDC

Accuracy

Technician Information

Date:

Section:

Team #:

Members:

DO NOT CHANGE/EDIT THE SHEET NAME

Calibration Station-Manometer as Standard

Pressure Port:

1

Manometer:

Oil

Mercury

Manometer Type:

U-tube

U-tube

Specific Gravity:

0.827

13.55

Calibration Scale:

mmOil

mmHg

Calibration Span:

700mm

42mm

Calibration Uncertainty:

Test Point	Manometer scale mm oil	Manometer scale mm Hg	Output VDC
1	0	0	0.11
2	100	6	1.213
3	200	12	2.302
4	300	18	3.38
5	400	24	4.459
6	500	30	5.554
7	600	36	6.635
8	700	42	7.714
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			

Observations and Comments During Testing

IWC vs Voltage

Applied Pressure (w.c.)	Transducer Output (VDC)
0	0.11
3.33	1.213
6.67	2.302
10	3.38
13.33	4.459
16.67	5.554
20	6.635
23.33	7.714

[illegible]